Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN · WOODEN BODY

CIE/666

Hand-driven insulation tester in seasoned wood

Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.



CIE/666 · ILLUSTRATIVE IMAGE

32	7	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

100 V · 250 V · 500 V · 1,000 V · 2,500 V · 5,000 V · 10,000 V

01 HIGHLIGHTS

- ✓ Seasoned wooden body with sunmica finish, detachable lid, carrying handle and lock
- ✓ Centrifugal clutch (governor) for constant output voltage
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Highly sensitive cross-coil indicator
- ✓ Accurate and quick response (3 seconds)
- ✓ Conforms to IS 2992:1987
- ✓ Silent generator
- ✓ Collapsible driven handle, folds back flush into the casing
- ✓ Wide anti-parallax mirror scale
- ✓ Long life and strength against vibration and impact
- ✓ Guard terminal and externally accessible infinity adjuster on ranges of 1000 MΩ and above

02 TYPICAL APPLICATIONS

- Motors, generators, and other rotating machinery
- Insulation control and switching equipment checks
- Distribution network and communication cable insulation testing
- HV switchgear and transformer insulation testing up to 10,000 V
- Locating intermittent shorts, defective connections, and developing insulation failures

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03 HOW IT WORKS

PRINCIPLE

A hand-driven clutch-type A.C. generator and a highly sensitive cross-coil indicator. The generated A.C. voltage is converted to D.C. by silicon diodes.

CONSTRUCTION

Housed in a sturdy, elegant seasoned wooden body with sunmica finish, a detachable lid, carrying handle and lock. The instrument is dust-free: rubber shoes hold the surface and give it stability.

04 SALIENT FEATURES

Silent generator

The A.C. generator runs extremely smoothly and silently thanks to the dynamic balance of its rotating parts, and is free from mechanical noise.

Centrifugal governor

A high-grade centrifugal governor holds the voltage constant for accurate measurement. Variation in crank speed does not affect the reading, and a cut-off clutch disengages the drive beyond the rated speed.

Collapsible handle

The generator handle folds back flush into the casing when the tester is not in use.

Quick response

3 seconds, when a resistance equal to the central scale mark is suddenly inserted between the measurement terminals, as per IS 2992:1987.

05 TECHNICAL SPECIFICATIONS

Body	Seasoned wooden with sunmica finish; detachable lid, carrying handle, and lock
Generator Type	Hand-driven, silent generator with centrifugal governor (clutch)
Scale Length	100 mm (approx.)
Accuracy	Conforms to IS: 2992-1987
Rotation Speed	160 r.p.m. or more
High Voltage Test	Withstands minimum 2000 V or (rated voltage + 1000) V A.C. (rms) between electrical circuit and handle, for 1 minute
Insulation Resistance Test	Better than 50 MΩ at 500 V between electrical circuit and containing case
Guard Terminal & Infinity Adjuster	Provided on instruments with insulation range 1000 MΩ and above
Response Time	3 seconds
Weight	4.1 kg (approx.)
Dimensions (L×W×H)	265 × 165 × 185 mm (approx.)

Ranges are listed on the following pages. Specifications are approximate where marked.

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06 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
100 V	0–20 · 0–50 · 0–100
250 V	0–50 · 0–100 · 0–250
500 V	0–100 · 0–200 · 0–500 · 0–1,000 · 0–2,000
1,000 V	0–100 · 0–200 · 0–500 · 0–1,000 · 0–2,000
2,500 V	0–1,000 · 0–2,000 · 0–2,500 · 0–5,000 · 0–10,000 · 0–20,000 · 50–50,000
5,000 V	0–5,000 · 0–10,000 · 0–20,000 · 50–50,000 · 100–100,000
10,000 V	0–10,000 · 0–20,000 · 50–50,000 · 100–100,000

Ranges shown are the standard builds. Any other range can be manufactured against specific requirements.

07 COMBINING VOLTAGES AND RANGES

Dual voltage & dual insulation range

One instrument, two rated voltages, each with its own insulation range, selected by a switch. The selected range must lie within the rated resistance listed for that voltage (voltage to measurable insulation range ratio up to 1:2 only; for any other ratio, enquire). Not available for the 10,000 V instrument.

Single voltage, dual insulation range

One rated voltage with two insulation ranges. Both ranges must lie within the rated resistance listed for that voltage. Not available for the 10,000 V instrument.

Continuity range (500 V and 1,000 V only)

Built in as an optional extra on single-voltage, single-range instruments, or on dual-voltage, dual-range instruments.

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08 MULTIRANGE SERIES: CIE/666/MR

SERIES	RATED VOLTAGE / INSULATION RANGE (MΩ)	CONTINUITY
MR-1	1,000 V / 0–2,000 500 V / 0–1,000 250 V / 0–500	0–20 kΩ
MR-2	1,000 V / 0–200 500 V / 0–100 250 V / 0–50	0–1 kΩ
MR-3	500 V / 0–1,000 250 V / 0–500 100 V / 0–200	0–5 kΩ
MR-4	500 V / 0–100 250 V / 0–50 100 V / 0–20	0–1 kΩ

SERIES	RATED VOLTAGE / INSULATION RANGE (MΩ)	CONTINUITY
MR-5	2,500 V / 0–5,000 1,000 V / 0–2,000 500 V / 0–1,000	0–20 kΩ
MR-6	2,500 V / 0–1,000 1,000 V / 0–200 500 V / 0–100	0–1 kΩ
MR-7	5,000 V / 0–20,000 2,500 V / 0–10,000	0–20 kΩ
MR-8	5,000 V / 0–5,000 2,500 V / 0–2,500	0–5 kΩ

Multirange instruments carry a range selector switch, so one instrument covers every voltage and range listed in its series. Anti-parallax mirror scale is not provided on triple-range and above instruments. Any other configuration can be built to order.

09 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case (wooden body)
- Testing leads, 1 m or 2 m (up to 1,000 V)
- H.V. testing leads, 2 m (general / special)

10 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/666, 500 V / 0–2,000 MΩ**. Any range not listed can be manufactured against specific requirements.

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